v16

Rotated joists / Doubled 2×10 girders

v16 CHANGES vs. v14

- Girders: 3 lines ⊥ house (y-dir) → 2 lines ∥ house (x-dir) at y=8'-0" and y=16'-0"
- Girder recipe: Tripled 2×10 (9 sticks) → Doubled 2×10 (fits Pylex 3.5" saddle)
- Pier count: 6 → 8 (4 per girder line @ 7'-8" spacing; within DCA-6 span limit)
- Joists: 32 segments ∥ house (x-dir) → 48 segments ⊥ house (y-dir) @ 12" o.c.
- Joist span: ~11' (v14) → 8' (v16) | Decking: ⊥ house → ∥ house (x-dir)

West Hartford Deck — v16 (Proposed) — Rotated Joists / Doubled 2×10 Girders

23'-6" × 16'-3" Overall Finished Surface (patio = 23'-6" × 16'-6")

MoistureShield Vision | Decking ∥ House | 4-Side Picture Frame

L-edge step: 42" wide @ 6'-0" | Far-edge step: 11'-9" @ right corner

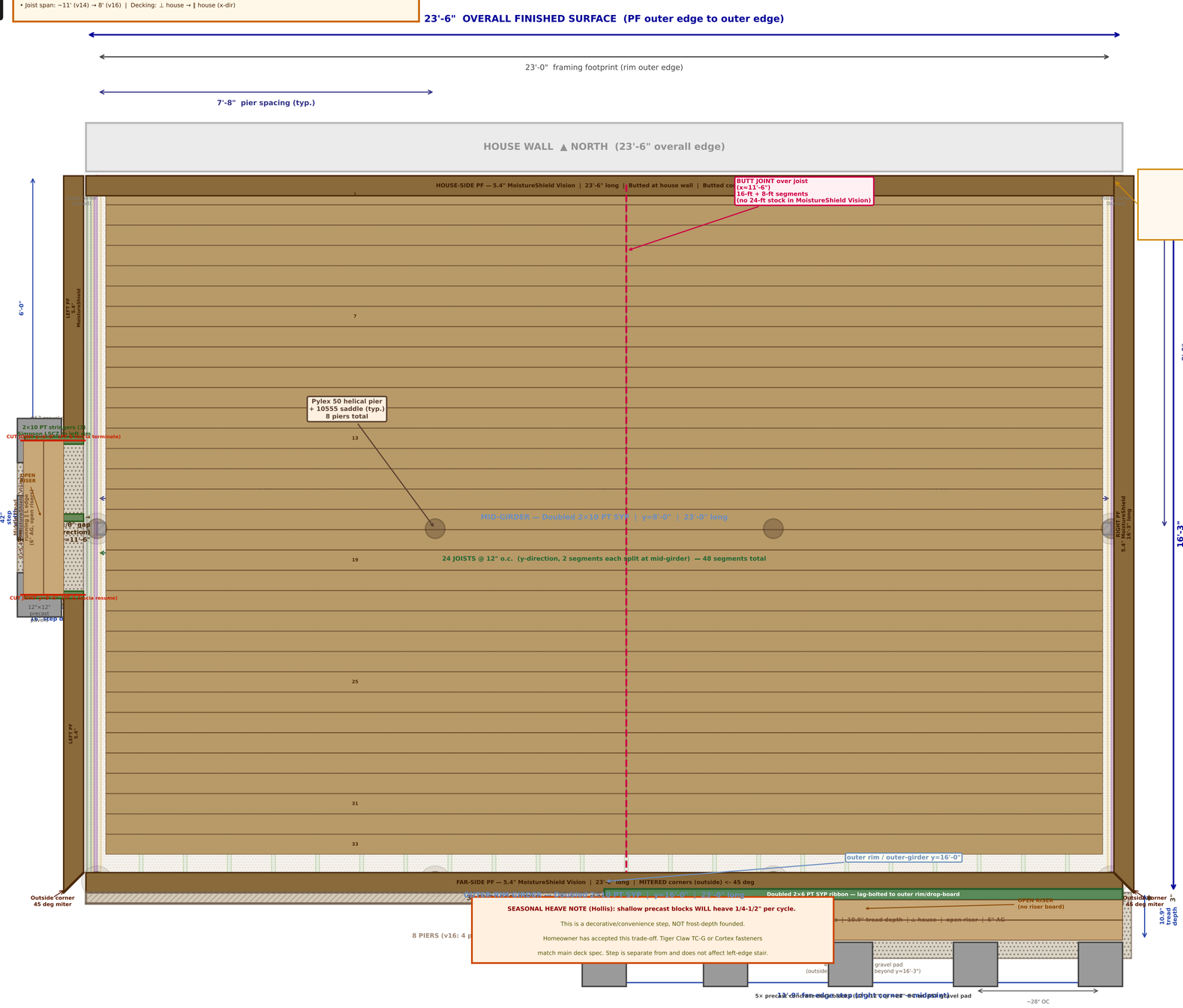
House on 23'-6" Edge NORTH

Scale: 1" = 1 ft (plot units = ft)

HOUSE = top edge

Date: 2026-05-09

Gemma — Technical Diagramming Specialist



LEDGER: ½" HDG carriage bolts, back-blocked from basement,
clustered at joist connections; ½" HDG lag at 18" OC two-row staggered
between connections (IRC R507.9.1.3). Wall = wood-frame, no masonry.
Basement access confirmed for back-blocking. Peel-and-stick + Z-flashing retained.

SEASONAL HEAVE NOTE (Hollis): shallow precast blocks WILL heave 1/4-1/2" per cycle.
This is a decorative/convenience step, NOT frost-depth founded.
Homeowner has accepted this trade-off. Tiger Claw TC-G or Cortex fasteners
match main deck spec. Step is separate from and does not affect left-edge stair.

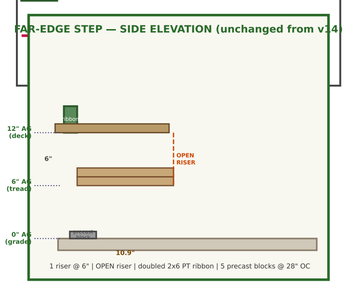
v16 FRAMING SUMMARY

Footprint:	23'-6" × 16'-3" overall (PF outer to PF outer)
Framing:	23'-0" × 16'-0"
Ledger:	Full-width at y=0, x-direction
Mid-girder:	Doubled 2×10 PT SYP | y=8'-0" | 23'-0" long
Outer-rim girder:	Doubled 2×10 PT SYP | y=16'-0" | 23'-0" long
Pier saddle:	Pylex 10555 (3.5" saddle) + shims — fits doubled 2×10
Piers:	8 total (4 per girder) @ x=0, 7'-8", 15'-4", 23'-0"
Pier span:	7'-8" (within DCA-6 limit of 8'-9" at 8'-0" joist span)
Joists:	24 × 2×10 PT SYP @ 12" o.c. | y-direction | 2 segments
Joist spans:	8'-0" each (seg A: ledger to mid-girder; seg B: mid to rim)
Decking:	MoistureShield Vision 5.4" | ∥ house | butt joint @ x=11'-6"
Stock:	16-ft + 8-ft segments (no 20-ft/24-ft in Vision line)

LEGEND — v16

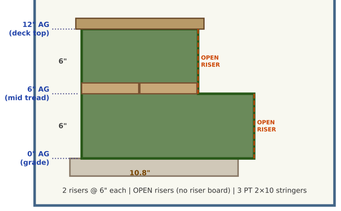
- **Field plank — 5.4" MoistureShield Vision (composite, capped)**
  Runs parallel to house (x-direction) | Butt joint @ x=11'-6" over joist
- **Picture-frame (PF) — 5.4" MoistureShield Vision, all 4 sides**
  L-side PF interrupted at L-edge step opening (y=6'-0" to y=9'-6")
- **Girder — Doubled 2×10 PT SYP | x-direction | 2 girder lines**
  Mid @ y=8'-0", Outer-rim @ y=16'-0" | Fits Pylex 3.5" saddle
- **Joist — 2×10 PT SYP | y-direction | 24 joists @ 12" o.c.**
  2 segments each | face-hung to ledger, mid-girder, outer-rim
- **Pylex 50 helical pier + 10555 saddle — 8 total**
  4 per girder line @ 7'-8" spacing | x=0, 7'-8", 15'-4", 23'-0"
- **L-edge stair stringer — 2×10 PT cut stringer (3 total)**
  Attached to left rim via Simpson LSCZ stringer hangers
- **Far-edge step ribbon — doubled 2×6 PT SYP**
  Lag-bolted to outer rim face; 11'-9" long, right corner to midpoint

FAR-EDGE STEP — SIDE ELEVATION (unchanged from v14)



1 riser @ 6" | OPEN riser | doubled 2×6 PT ribbon | 5 precast blocks @ 28" OC

LEFT-EDGE STAIR — SIDE ELEVATION (unchanged from v14)



2 risers @ 6" each | OPEN risers (no riser board) | 3 PT 2×10 stringers